VIVEK SHRAVAN GUPTA

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AI/ML-focused Software Engineer with hands-on experience in **human-in-the-loop AI systems**, **LLM evaluation**, and applied machine learning. Experienced in improving model quality through structured prompt design, expert review, error analysis, and benchmarking. Strong background in ML-driven data platforms, scalable pipelines, and cloud-based systems.

WORK EXPERIENCE

AI Systems Evaluation Engineer | Handshake – Chicago, USA (Remote) Sep 2025 – Present

- Developed and evaluated domain-specific prompts to assess model performance, reasoning ability, and consistency of **large language models (LLMs)** in a human-in-the-loop evaluation setting.
- Analyzed LLM outputs for accuracy, clarity, logical structure, and depth, identifying gaps and patterns to support systematic **model evaluation**.
- Provided expert reviews and structured feedback to improve AI understanding of complex concepts and nuanced queries.
- Conducted independent research and background analysis to support prompt design, evaluation criteria, and benchmarking workflows.
- Supported iterative model improvement workflows by tracking evaluation outcomes and feedback trends across experiments.

Research Assistant | DePaul University – Chicago, USA Feb 2024 – Jun 2025

- Designed and built a platform enabling researchers to work efficiently with TB-scale data and **machine learning models**, significantly accelerating data analysis.
- Improved data processing speed by 52% using ML techniques including clustering, regression, and dimensionality reduction.
- Led development of a **CI/CD pipeline** using Django and Python (Pandas, NumPy, Scikit-learn) to clean, organize, and manage large datasets.
- Collaborated with data scientists and engineers to deliver scalable machine learning solutions for academic and industry use cases.

Full Stack Developer | Laxmi Print-o-Pack – Mumbai, India Dec 2021 – May 2022

- Developed and maintained full-stack web applications for multiple clients using React.js, Node.js, Express.js, and MongoDB, reducing development time by 34% through efficient implementation and code optimization.
- Built responsive and intuitive user interfaces with React.js and Tailwind CSS, increasing user engagement by 28% across desktop, tablet, and mobile platforms.
- Designed and integrated **RESTful APIs**, ensuring seamless communication between frontend and backend systems while improving application scalability.
- Created responsive and user-friendly interfaces with **React.js** and **Tailwind CSS**, making sure websites work smoothly on desktop, tablet, and mobile. This led to a 28% increase in user engagement.

PROJECTS

AI-Based Customer Support Chatbot

- Developed an **AI-powered chatbot** using Google Cloud NLP, **TensorFlow**, and **NLP** techniques, reducing customer service response time by 40% and improving customer satisfaction. Integrated with Django for dynamic **real-time interaction**.
- Configured backend on AWS-compatible architecture, utilizing MongoDB for storing interactions and implemented Socket.io for seamless real-time chat, handling over 1,000 requests daily.

Destiny - Campus Virtual Tour and Navigation

- Led development of a **Unity3D**-based immersive campus tour using **Android SDK**, ARCore, and GPS, enhancing navigation by 30% through real-time geo-location features. Configured system's cloud-readiness using GCP & AWS.
- Designed front-end with C#, Java and Kotlin for responsive, cross-platform support using JavaScript and SQL for backend

NeuroBazaar - Interactive ML & Data Visualization Platform

- Independently designed and developed the complete web portal for an interactive Machine Learning and Data Science Visualization Platform, owning the **end-to-end development** of the user-facing application.
- Built the portal experience to make ML and data-oriented workflows accessible through an **interactive web interface**; managed feature development, UI improvements, debugging, and ongoing maintenance.

TECHNICAL SKILLS

Programming Languages:	Java, Python, JavaScript, C, C++, C#, R, Swift, Kotlin.
Web Technologies:	HTML, CSS, PHP, JavaScript, React.js, Vue.js, Tailwind, Bootstrap, jQuery, AJAX.
Frameworks:	Django, Flask, Node.js, React.js, Unity3D, AR Core, .NET, Android SDK, iOS SDK.
Databases:	SQL, MySQL, PostgreSQL, MongoDB, SQLite, Oracle.
Cloud Technologies:	Chameleon Cloud, AWS, Google Cloud Platform, Microsoft Azure.
AI / ML Concepts:	Model Evaluation, Human-in-the-Loop Systems, Prompt Engineering, Experimentation & Benchmarking, Error Analysis, Data Validation, ML Pipelines.

EDUCATION

DePaul University – Chicago, IL | Master of Science in Computer Science (**CGPA**: **3.95/4**) Sep 2023 – Jun 2025

University Of Mumbai – Mumbai, IN | Bachelor of Engineering in Computer Engineering (**CGPA**: **9.43/10**) Jun 2020 – May 2023